Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | | | | | |

Candidate Number | | | | |

Pearson Edexcel International Advanced Level

Time 1 hour 20 minutes

Paper reference **WCH13/01**

Chemistry

International Advanced Subsidiary/Advanced Level
UNIT 3: Practical Skills in Chemistry I

You must have:
Scientific calculator, ruler

Total Marks

Instructions

- Use **black** ink or **black** ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided – *there may be more space than you need.*

Information

- The total mark for this paper is 50.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- You will be assessed on your ability to organise and present information, ideas, descriptions and arguments clearly and logically, including your use of grammar, punctuation and spelling.
- A Periodic Table is printed on the back cover of this paper.

Advice

- Read each question carefully before you start to answer it.
- Show all your working in calculations and include units where appropriate.
- Check your answers if you have time at the end.

Turn over

Pearson

Answer ALL the questions. Write your answers in the spaces provided.

1 This question is about ammonium chloride, NH_4Cl, a soluble ionic compound.

(a) An aqueous solution of NH_4Cl contains both ammonium ions, NH_4^+, and chloride ions, Cl^-.

(i) State what would be **seen** on the addition of acidified silver nitrate solution to an aqueous solution of NH_4Cl.

(1)

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(ii) Describe a test to confirm the presence of NH_4^+ ions in a solution of NH_4Cl. Include the result of the positive test.

(2)

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(b) A student investigated the enthalpy change when dissolving NH_4Cl in excess water.

$$NH_4Cl(s) + aq \rightarrow NH_4^+(aq) + Cl^-(aq)$$

Procedure

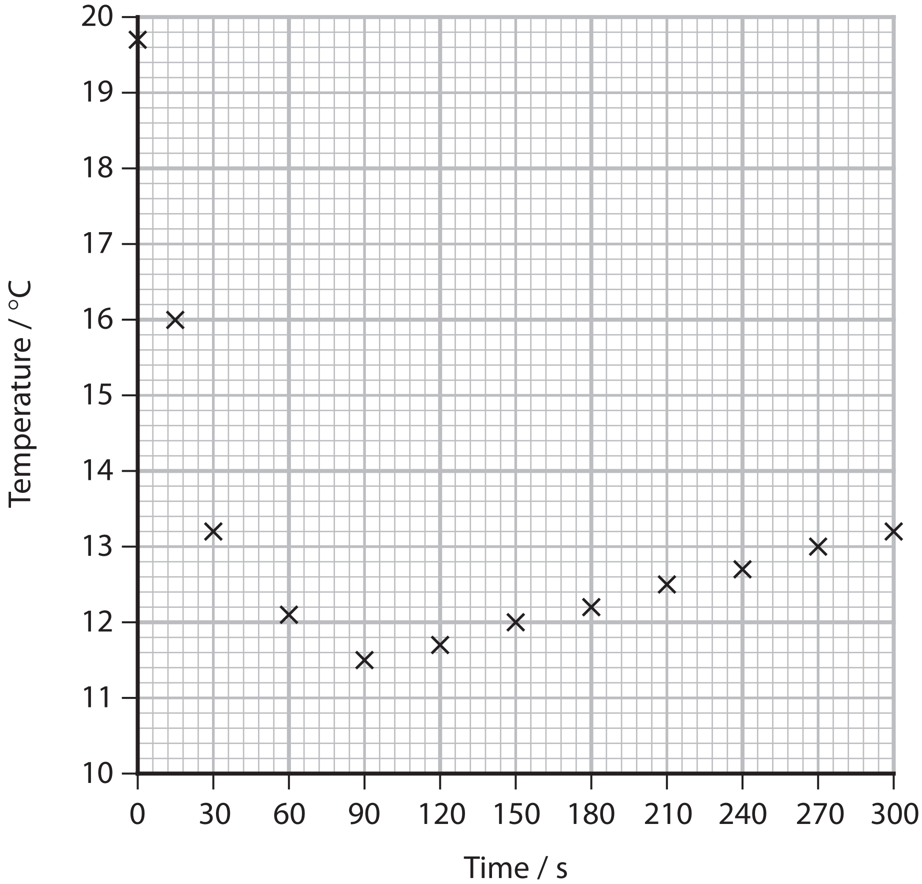
Step **1** Accurately weigh 7.17 g of NH_4Cl into a glass beaker.

Step **2** Fill a 50 cm^3 measuring cylinder with deionised water. Measure the temperature of the water using a thermometer.

Step **3** Pour the water from the measuring cylinder into the beaker and at the same time start a stopwatch. Stir the solution in the beaker, using the thermometer.

Step **4** Record the temperature at 15 s, 30 s and then at 30 s intervals while continuing to stir the solution.

The data from the experiment are shown on the graph.



(i) Give **two** reasons why the student stirred the solution in Steps **3** and **4**.

(2)

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(ii) Use the graph to determine the maximum temperature change, ΔT, in this experiment. You **must** show your working on the graph.

(2)

(iii) Another student carried out the experiment using a polystyrene cup in place of the glass beaker.

Explain how this student's graph would be different.
You may annotate the graph as part of your answer.

(3)

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(c) The experimental results of another student were used in the equation shown to calculate the enthalpy change, ΔH, for dissolving one mole of NH_4Cl in excess water.

$$\Delta H = \frac{m \times c \times \Delta T}{n}$$

$$= +14\,500\ \text{J mol}^{-1}$$

In the equation

m = mass of solution = 50 g

c = specific heat capacity of water = 4.18 J g^{-1} °C^{-1}

ΔT = maximum temperature change of solution

n = moles of NH_4Cl

(i) State **two** assumptions made in this calculation.
You do **not** need to justify your answers.

(2)

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(ii) The total percentage uncertainty in this experiment was 2.6 %.

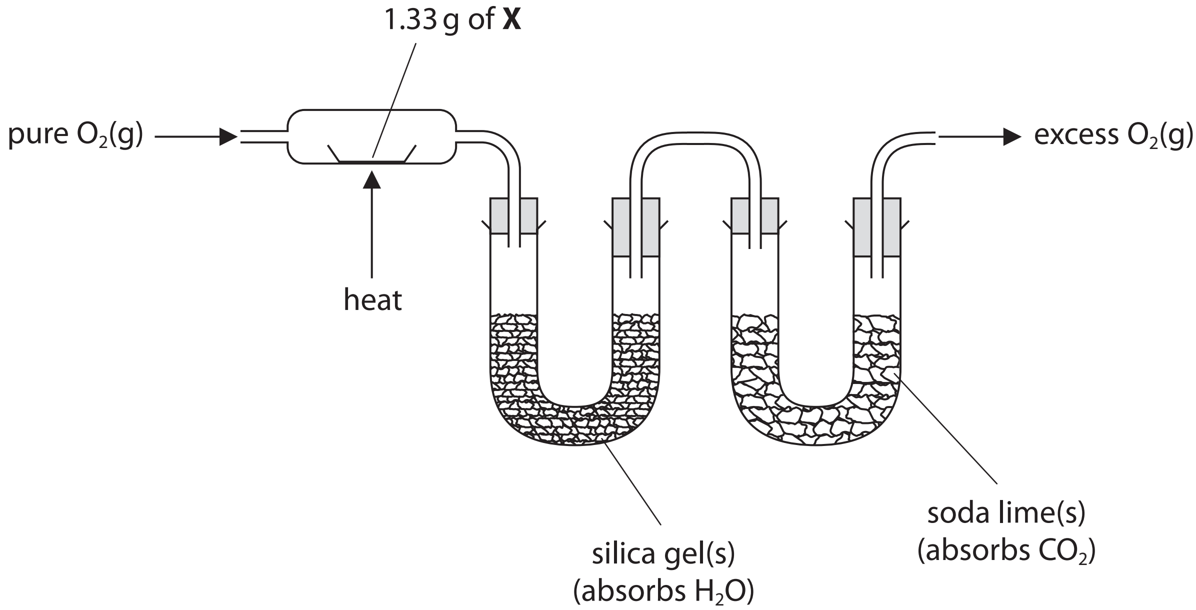
Show that the enthalpy change of 14.5 $kJ\,mol^{-1}$ is consistent with a data book value of 14.8 $kJ\,mol^{-1}$.

(2)

(Total for Question 1 = 14 marks)

2 This question is about two organic compounds, **X** and **Y**. Both are liquids which contain carbon, hydrogen and oxygen only.

(a) The mass of hydrogen and of carbon present in 1.33 g of **X** were determined by passing its combustion products through the apparatus shown.



(i) State the **measurements** that should be made.

(2)

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(ii) Give **two** reasons why pure O_2(g), and **not** air, should be used.

(2)

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(iii) The experiment showed that 1.33 g of **X** contains 0.14 g of hydrogen and 0.63 g of carbon.

Calculate the empirical formula of **X**, using these data.
You **must** show your working.

(3)

(b) When phosphorus(V) chloride is added to **X**, steamy white fumes are seen.

State what can be deduced about compound **X** from this observation only.

(1)

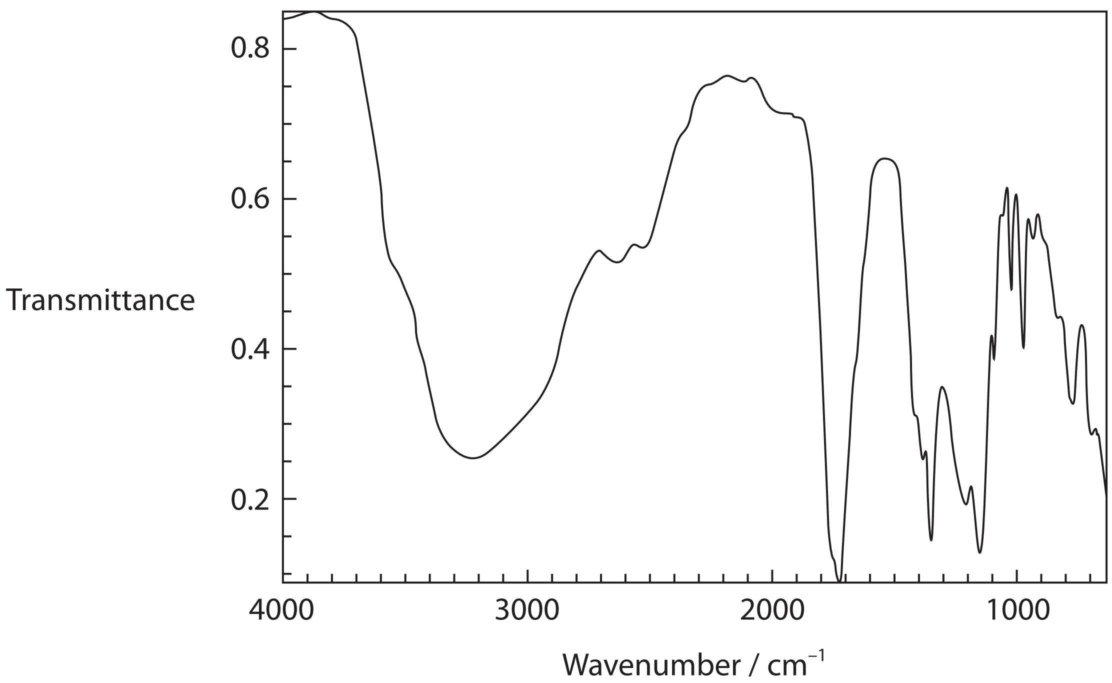
(c) Compound **X** is converted into compound **Y** when refluxed with **excess** sodium dichromate(VI) in sulfuric acid.

Compound **Y** is a liquid that is soluble in the reaction mixture.

Draw a **labelled** diagram of the apparatus that could be used to separate **Y** from the reaction mixture.

(3)

(d) The infrared spectrum of **Y** is shown.



The table shows some infrared absorption data.

Bond	Wavenumber range / cm^{-1}
C—H (alkane)	2962 – 2853
O—H (alcohols and phenols)	3750 – 3200
O—H (carboxylic acids)	3300 – 2500
C═C (alkene)	1669 – 1645
C═O (aldehydes, ketones, carboxylic acids)	1740 – 1680

Explain how this spectrum shows that **Y** contains a carboxylic acid functional group, quoting data from the table.

(2)

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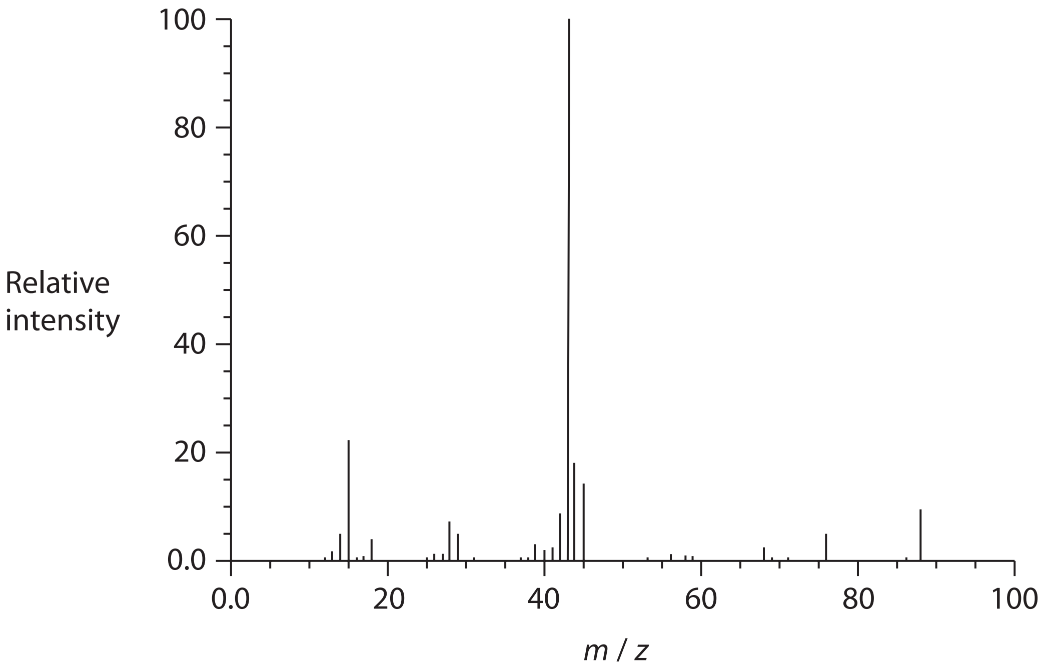
(e) The mass spectrum of **Y** is shown.



(i) Show that the mass spectrum is consistent with **Y** having the molecular formula $C_3H_4O_3$.

(1)

(ii) Suggest the structure of the ion causing the peak at $m / z = 43$ in the mass spectrum of **Y**.

(1)

(f) Compound **X** contains one type of functional group.

Compound **Y** contains two different functional groups.

Use the information in the question to deduce the structures of **X** and **Y**.

(2)

Compound **X**

Compound **Y**

(Total for Question 2 = 17 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

3 A student used a precipitation titration to determine the value of **x** in the formula of a sample of hydrated barium chloride, $BaCl_2 \cdot \mathbf{x}H_2O$.

Procedure

Step **1** Prepare a solution by dissolving 1.57 g of $BaCl_2 \cdot \mathbf{x}H_2O$ in deionised water, making the solution up to the mark in a 250.0 cm^3 volumetric flask and then mixing thoroughly.

Step **2** Use a pipette to transfer 10.0 cm^3 of the barium chloride solution into a conical flask.
Add excess sodium sulfate solution and swirl the mixture.

Step **3** Fill a burette with 0.0324 mol dm^{-3} silver nitrate solution.

Step **4** Add three drops of potassium chromate(VI) solution to the conical flask and titrate the contents, while swirling, with the silver nitrate solution.
The end-point is shown by the appearance of a permanent pale red precipitate.

Step **5** Repeat Steps **2** to **4** until concordant results are obtained.

During the titration, two precipitation reactions occur.

Reaction **1** Silver ions react with chloride ions forming silver chloride.

$$Ag^+(aq) \ + \ Cl^-(aq) \ \rightarrow \ AgCl(s)$$

Reaction **2** Once all chloride ions have reacted, silver ions react with chromate(VI) ions to form a red precipitate of silver chromate(VI).

$$2Ag^+(aq) \ + \ CrO_4^{2-}(aq) \ \rightarrow \ Ag_2CrO_4(s)$$

(a) (i) Give the **ionic** equation for the reaction that occurs when sodium sulfate solution is added to the conical flask in Step **2**.
Include state symbols.

(1)

(ii) Give a possible reason why it is necessary to add sodium sulfate solution.
Justify your answer.

(1)

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(b) Suggest why the red precipitate of silver chromate(VI) only forms after all the chloride ions have reacted.

(1)

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(c) Some data obtained in the experiment are shown.

Titration number	1	2	3	4
Burette reading (final) / cm^3	16.15	32.05	48.30	47.40
Burette reading (initial) / cm^3	0.00	16.15	32.50	31.55
Titre / cm^3	16.15			

(i) Complete the table and use the concordant results to calculate the mean titre.

(2)

(ii) Determine the value of **x** in the formula of the hydrated salt, $BaCl_2 \cdot \mathbf{x}H_2O$.
Use information from the procedure and your mean titre from (c)(i).
You **must** show your working.

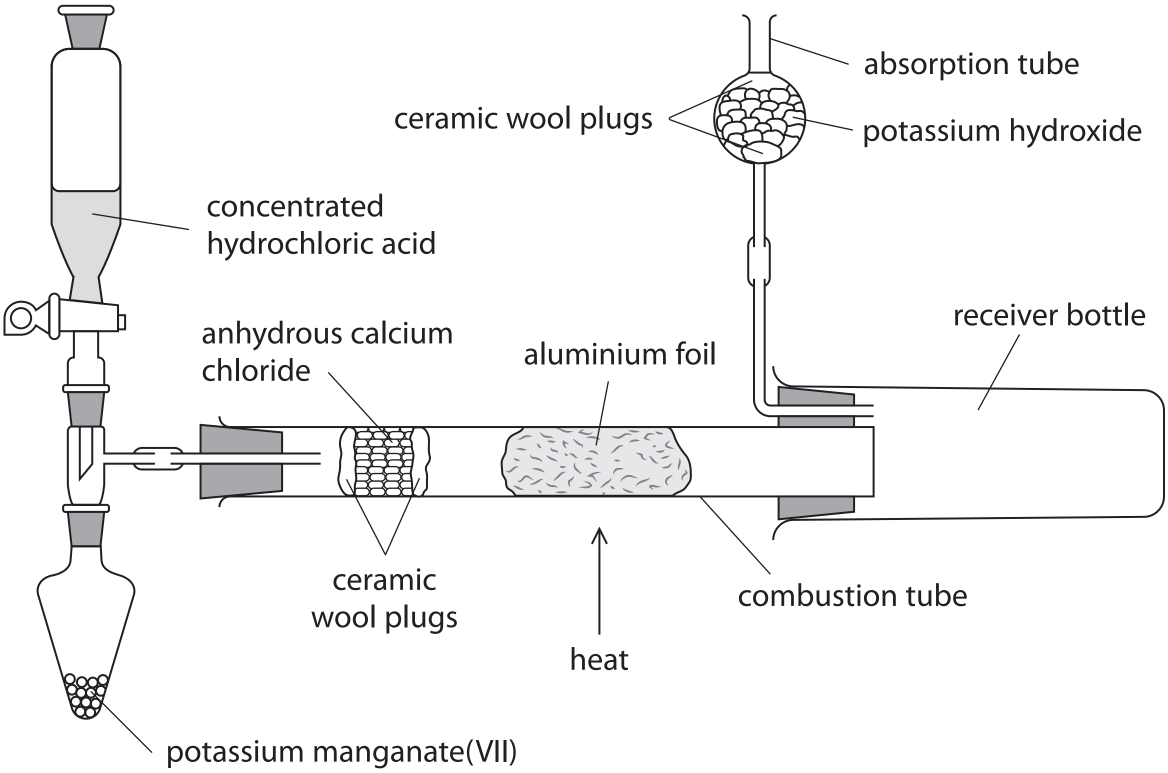
(5)

(Total for Question 3 = 10 marks)

4 This question is about the preparation of anhydrous aluminium chloride, $AlCl_3$, which reacts vigorously with water and must be stored in tightly sealed containers.

A sample of anhydrous $AlCl_3$ was prepared by passing chlorine gas over hot aluminium foil using the apparatus shown.

$$2Al(s) + 3Cl_2(g) \rightarrow 2AlCl_3(s)$$



Procedure

Step **1** Assemble the apparatus with about 5 g of potassium manganate(VII) in the pear-shaped flask, 10 cm^3 of concentrated hydrochloric acid in the tap funnel and a known mass of aluminium foil in the combustion tube.

Step **2** Carefully open the tap of the funnel, allowing the acid to enter the pear-shaped flask drop by drop. Wait for twenty seconds.

Step **3** Heat the aluminium foil until it glows brightly. Continue heating until the reaction is complete. Allow the apparatus to cool before closing the tap of the funnel.

Step **4** Remove the receiver bottle, quickly scrape the product into a sample tube and seal with a lid.

P 6 7 1 2 9 A 0 1 6 2 0

(a) Granules of anhydrous calcium chloride are held between two ceramic wool plugs in the combustion tube.

(i) Explain the purpose of the anhydrous calcium chloride.

(2)

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(ii) Give the reason why granules of anhydrous calcium chloride are used rather than powder.

(1)

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(b) The reaction occurring in Step **2** produces chlorine gas.

(i) Identify the main hazard related to chlorine gas, giving the **best** way of minimising the risk when using this gas.

(2)

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(ii) Give a reason why the concentrated hydrochloric acid is added 'drop by drop' to the pear-shaped flask.

(1)

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(c) Suggest why the heating of the aluminium in Step **3** is delayed by 20 s after the initial production of chlorine gas.

(1)

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(d) State how you would know the reaction is complete in Step **3**.

(1)

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(e) Suggest the purpose of the potassium hydroxide in the absorption tube.

(1)

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(Total for Question 4 = 9 marks)

TOTAL FOR PAPER = 50 MARKS

The Periodic Table of Elements

Key

relative atomic mass **atomic symbol** name atomic (proton) number

1 (1)	2 (2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	3 (13)	4 (14)	5 (15)	6 (16)	7 (17)	0 (8) (18)
							1.0 **H** hydrogen 1										4.0 **He** helium 2
6.9 **Li** lithium 3	9.0 **Be** beryllium 4											10.8 **B** boron 5	12.0 **C** carbon 6	14.0 **N** nitrogen 7	16.0 **O** oxygen 8	19.0 **F** fluorine 9	20.2 **Ne** neon 10
23.0 **Na** sodium 11	24.3 **Mg** magnesium 12											27.0 **Al** aluminium 13	28.1 **Si** silicon 14	31.0 **P** phosphorus 15	32.1 **S** sulfur 16	35.5 **Cl** chlorine 17	39.9 **Ar** argon 18
39.1 **K** potassium 19	40.1 **Ca** calcium 20	45.0 **Sc** scandium 21	47.9 **Ti** titanium 22	50.9 **V** vanadium 23	52.0 **Cr** chromium 24	54.9 **Mn** manganese 25	55.8 **Fe** iron 26	58.9 **Co** cobalt 27	58.7 **Ni** nickel 28	63.5 **Cu** copper 29	65.4 **Zn** zinc 30	69.7 **Ga** gallium 31	72.6 **Ge** germanium 32	74.9 **As** arsenic 33	79.0 **Se** selenium 34	79.9 **Br** bromine 35	83.8 **Kr** krypton 36
85.5 **Rb** rubidium 37	87.6 **Sr** strontium 38	88.9 **Y** yttrium 39	91.2 **Zr** zirconium 40	92.9 **Nb** niobium 41	95.9 **Mo** molybdenum 42	[98] **Tc** technetium 43	101.1 **Ru** ruthenium 44	102.9 **Rh** rhodium 45	106.4 **Pd** palladium 46	107.9 **Ag** silver 47	112.4 **Cd** cadmium 48	114.8 **In** indium 49	118.7 **Sn** tin 50	121.8 **Sb** antimony 51	127.6 **Te** tellurium 52	126.9 **I** iodine 53	131.3 **Xe** xenon 54
132.9 **Cs** caesium 55	137.3 **Ba** barium 56	138.9 **La*** lanthanum 57	178.5 **Hf** hafnium 72	180.9 **Ta** tantalum 73	183.8 **W** tungsten 74	186.2 **Re** rhenium 75	190.2 **Os** osmium 76	192.2 **Ir** iridium 77	195.1 **Pt** platinum 78	197.0 **Au** gold 79	200.6 **Hg** mercury 80	204.4 **Tl** thallium 81	207.2 **Pb** lead 82	209.0 **Bi** bismuth 83	[209] **Po** polonium 84	[210] **At** astatine 85	[222] **Rn** radon 86
[223] **Fr** francium 87	[226] **Ra** radium 88	[227] **Ac*** actinium 89	[261] **Rf** rutherfordium 104	[262] **Db** dubnium 105	[266] **Sg** seaborgium 106	[264] **Bh** bohrium 107	[277] **Hs** hassium 108	[268] **Mt** meitnerium 109	[271] **Ds** darmstadtium 110	[272] **Rg** roentgenium 111							

Elements with atomic numbers 112-116 have been reported but not fully authenticated

140 **Ce** cerium 58	141 **Pr** praseodymium 59	144 **Nd** neodymium 60	[147] **Pm** promethium 61	150 **Sm** samarium 62	152 **Eu** europium 63	157 **Gd** gadolinium 64	159 **Tb** terbium 65	163 **Dy** dysprosium 66	165 **Ho** holmium 67	167 **Er** erbium 68	169 **Tm** thulium 69	173 **Yb** ytterbium 70	175 **Lu** lutetium 71
232 **Th** thorium 90	[231] **Pa** protactinium 91	238 **U** uranium 92	[237] **Np** neptunium 93	[242] **Pu** plutonium 94	[243] **Am** americium 95	[247] **Cm** curium 96	[245] **Bk** berkelium 97	[251] **Cf** californium 98	[254] **Es** einsteinium 99	[253] **Fm** fermium 100	[256] **Md** mendelevium 101	[254] **No** nobelium 102	[257] **Lr** lawrencium 103

* Lanthanide series

* Actinide series